

Uniform quasipolynomial-time trace reconstruction

OpenAI

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Abstract

We give a uniform algorithm that reconstructs every binary string from independent deletion traces when its length and rational retention probability are known. For each fixed retention probability, both the number of traces and the bit complexity are quasipolynomial in the string length. More generally, we give an explicit sample bound uniform over all rational retention probabilities, with running time polynomial in the sample budget and the binary input length. If the deletion probability is at most $n^{-\varepsilon}$ for fixed $\varepsilon > 0$, the sample and running-time bounds are polynomial. Reconstruction succeeds with probability at least $2/3$ for each input string.

1 Introduction

A deletion trace of a binary string is obtained by retaining each position independently with probability p and concatenating the retained bits in order. The trace contains its length and its bits, but does not identify their original positions. Trace reconstruction asks for the original string from independent repetitions of this experiment. We consider the worst case: the unknown string is arbitrary, and the recovery probability must hold separately for each string.

Knowing that a collection of statistics distinguishes every pair of strings gives a sample bound, but an efficient decoder needs a way to find the unknown string. We obtain such a decoder by extending separation to a convex family of moment arrays. The resulting tests determine one bit at a time and can be implemented using rational arithmetic.

1.1 The uniform guarantee

Let $n \geq 2$ be the known original length and let $p = a/b \in (0, 1]$ be known, where a, b are positive integers in lowest terms. Write L for the total binary length of a and b , and put $q = 1 - p$. All logarithms are natural unless a base is displayed. Define

$$\begin{aligned} X(n, p) &= \begin{cases} 1 + \frac{\log n}{1 + \log(1/q)}, & q > 0, \\ 1, & q = 0, \end{cases} \\ B(n, p) &= p^{-1}(\log n)X(n, p)^2(1 + \log X(n, p))^6. \end{aligned} \tag{1.1}$$

Theorem 1.1 (Uniform trace reconstruction). *There exist absolute constants $C, c \geq 1$ and one randomized multitape Turing machine $\mathcal{A}$ with the following property. For every $n \geq 2$, every rational $p \in (0, 1]$, and every $x \in \{0, 1\}^n$, given n, p in binary and access to independent ordinary deletion traces of x , the machine requests at most*

$$M_C(n, p) = \lceil \exp(CB(n, p)) \rceil$$

traces, outputs an element of $\{0,1\}^n$, and returns x with probability at least $2/3$.

For every internal random-bit sequence and every well-formed trace stream of strings of length at most n , it halts within

$$(n + L + M_C(n, p))^c$$

bit operations. This bound includes preprocessing, all parameter and finite-precision calculations, every received trace bit and delimiter, and the output. The only oracle operation is the external generation of the independent deletion traces.

For fixed p , the budget in (1.1) is $O_p((\log n)^3(1 + \log \log n)^6)$ as $n \rightarrow \infty$. Thus both resources in Theorem 1.1 are quasipolynomial. The theorem also controls channels whose retention probability varies with n , including probabilities arbitrarily close to zero or one. Its running-time guarantee is polynomial in the permitted sample budget. For every fixed $\kappa > 0$, the regime $q \leq n^{-\kappa}$ has polynomial sample complexity in n and polynomial bit complexity in $n + L$: indeed, $X(n, p) \leq 1 + 1/\kappa$ and p^{-1} is bounded by a constant depending only on κ , so $B(n, p) = O_\kappa(\log n)$.

1.2 History and relation to earlier work

Reconstructing a sequence from corrupted copies has a broad information-theoretic history; Levenshtein studied reconstruction under several combinatorial and probabilistic channel models [15]. Batu, Kannan, Khanna, and McGregor formulated reconstruction from independent random deletion traces in theoretical computer science, motivated in part by sequence alignment [2]. The missing positions make the problem different from coordinatewise denoising: the same trace rank can originate at many input positions.

For a fixed deletion probability, Holenstein, Mitzenmacher, Panigrahy, and Wieder obtained a worst-case sample bound $\exp(\tilde{O}(\sqrt{n}))$ [13]. In independent papers published at STOC 2017, De, O'Donnell, and Servedio, and Nazarov and Peres, obtained $\exp(O(n^{1/3}))$ bounds and matching limitations for methods using only trace means [11, 17]. Their complex-analytic approach connects deletion statistics to bounded-coefficient polynomials; analytic inequalities of Borwein and Erdélyi are part of the background to this method [3]. The mean-based lower bound is a restriction on that method, not on arbitrary trace algorithms.

Chase improved the general sample upper bound to $\exp(O(n^{1/5} \log^5 n))$ [6]. On the lower-bound side, Holden and Lyons proved an almost- $n^{5/4}$ bound, and Chase strengthened it to $\Omega(n^{3/2}/\log^7 n)$ for fixed deletion probability [12, 5]. Burudgunte, Valiant, and Wang subsequently proved the sample bound

$$\exp(p^{-7/3}(\log_2 n)^{c_0})$$

for an absolute c_0 , using nonlocal low-order statistics and a multiscale propagation argument [4, Theorem 43]. This bound is quasipolynomial when p^{-1} is polylogarithmic in n . Theorem 1.1 builds on this strategy and provides an explicit uniform reconstruction algorithm with the displayed bit-complexity bound.

Sample complexity and reconstruction time require different arguments. At fixed confidence, maximum likelihood achieves worst-case reconstruction using at most $O(n)$ times the optimal sample count, but evaluating it over all candidate strings entails a large search [10, Corollary 1.1 of the full version]. Burudgunte, Valiant, and Wang explicitly distinguish testing two known strings from reconstructing an unknown string, and leave quasipolynomial-time reconstruction open for their approach [4, Section 1]. The additional separation proved here concerns every positive semidefinite moment completion of a wrong prefix. It enables constructive feasibility tests in place of a search over complete strings. Earlier polynomial-time results already cover low-deletion regimes [8].

The use of nonlocal statistics is essential to the comparison with known barriers. Chen, De, Lee, and Servedio proved strong lower bounds for statistical queries restricted to short *consecutive* trace blocks [7]. Our statistics involve few ranks but may span the whole trace, as in [4]. The generating-function identity used to transfer them from input coordinates has antecedents in higher-order trace reconstruction [9, 6].

The positive moment relaxation is a Boolean instance of the semidefinite moment and sum-of-squares framework [14, 18]. The finite feasibility iteration is related to the classical relaxation methods of Agmon and Motzkin–Schoenberg [1, 16]. We prove the analytic separation and the rational implementation directly; no external trace-reconstruction or optimization theorem is needed.

1.3 Proof strategy and additional ingredients

The decoder estimates products of bits at specified trace ranks; those ranks may be far apart. It then tests a proposed prefix by asking for moments of the remaining input bits that agree with these estimates within a prescribed tolerance. The moment array is constrained so that the expected squared magnitude of every polynomial up to a prescribed degree is nonnegative. Actual distributions give such arrays, but an arbitrary feasible array need not come from a distribution.

The central statement separates the true input from every such array fixing a wrong prefix. Let d be the first wrong position, and write $\Delta[P]$ for the difference between the array’s value on a polynomial P and its value at the true input. Then $\Delta[X_d] = \pm 1$, whereas all differences strictly before d vanish. We propagate this local signal through progressively larger distance scales. At each stage, the statistics are weighted sums over increasing tuples of input positions, with exponential weights on their gaps. We track an exponentially damped Fourier sum of the discrepancies of these tuple statistics, indexed by the last input position. The aim is to reduce its damping and frequency while retaining a detectable discrepancy.

Two features make this propagation compatible with the convex relaxation. First, positivity turns polynomial moments into inner products. If squared norms and neighboring inner products along a chain of J vectors differ from one by at most ξ , then a telescoping argument bounds the endpoint error by $O(J^2\xi)$. Applied to normalized Fourier sums and their products, this forces a detectable product whose total frequency is small. Pairing frequencies at prescribed differences under a Cauchy density supplies enough pairs for this chain.

Second, a Gaussian contour deformation converts each such product back into a tuple statistic with smaller decay and frequency. The paths joining consecutive positions in each old tuple, together with links between their endpoints, form a connected graph. A spanning tree assigns distinct edges to the new gaps, controlling the coefficient mass under interleaving. This retains the signal while the tuple order increases by a factor of at most four.

After the last scale, a generating-function identity transfers input statistics to ordinary trace statistics. In this identity, the channel substitution $z = q + pw$ maps the unit disk in a trace gap parameter w to the disk $|z - q| \leq p$ in the corresponding input parameter z . A single complex strip moves all input gap parameters into these disks simultaneously. This supplies the quantitative separation theorem without requiring the algorithm to search for any of the analytic witnesses.

Finally, when the empirical estimates are sufficiently accurate, a true prefix has a moment completion with strict slack: give independent uniform signs a sufficiently small weight in a mixture with its true completion. This preserves the statistical tolerance while making the moment matrix positive definite. We exploit that slack with an explicit sequence of rational updates, exact positivity tests, and rounding to a fixed dyadic grid. A fixed iteration cap gives the worst-stream bound in Theorem 1.1.

Section 2 introduces the relaxation. Section 3 proves the multiscale step, and Section 4 transfers its signal to traces. Section 5 constructs the decoder and proves Theorem 1.1.

2 Moment relaxations and prefix separation

We first describe the finite collection of statistics used by the decoder and the convex set against which they will separate the true input. Throughout, $[n] = \{1, \dots, n\}$ and $X_1, \dots, X_n$ are formal Boolean variables. Products and degrees are understood after the reductions $X_i^2 = X_i$.

2.1 Observable trace moments

For a nonempty rank set $I = \{\ell_1 < \dots < \ell_k\} \subseteq [n]$, the corresponding trace observable is the product of the bits at those ranks, with value zero if the trace has length less than ℓ_k . Let $T_I(X)$ be its channel expectation, regarded as a polynomial in the input bits. More explicitly,

$$T_I(X) = \sum_{i_1 < \dots < i_k} c_{I; i_1, \dots, i_k} X_{i_1} \cdots X_{i_k},$$

where $c_{I; i_1, \dots, i_k}$ is the probability that the selected input positions appear at ranks $\ell_1, \dots, \ell_k$ in a deletion trace. These events are disjoint as the input tuple varies. Consequently the coefficients are nonnegative, their sum is at most one, and the degree is at most k . They are rational when p is rational. The algorithm will estimate $T_I(x)$ by averaging the observable over independent traces; it never observes the input positions that determine the coefficients.

2.2 Positive moment arrays

The following Boolean moment relaxation is a specialization of the semidefinite moment and sum-of-squares frameworks of Lasserre [14, Section 2] and Parrilo [18, Sections 4.2–4.3]. We use only its defining positivity condition and elementary consequences, proved below. Fix a proposed prefix $u \in \{0, 1\}^d$ and an integer $D \geq 1$. On the remaining positions $U = \{d+1, \dots, n\}$, use sign variables $Y_i = 2X_i - 1$, so $Y_i^2 = 1$. A moment array consists of real numbers m_J for $J \subseteq U$ with $|J| \leq 2D$, including $m_\emptyset = 1$, subject to

$$\mathcal{M}_D(m) = (m_{I \triangle J})_{I, J \subseteq U, |I|, |J| \leq D} \succeq 0. \quad (2.1)$$

Only existing subsets are indexed; thus the definition also applies when $D > |U|$ or U is empty.

The array defines a functional $\mathcal{E}$ on polynomials of degree at most $2D$ by substituting $X_i = u_i$ for $i \leq d$, expanding the other bits as $(1 + Y_i)/2$, reducing $Y_i^2 = 1$, and setting $\mathcal{E}[\prod_{i \in J} Y_i] = m_J$. We extend this functional complex linearly. We call it a *positive moment functional of order $2D$ fixing u* . This terminology asserts (2.1), not the existence of a probability distribution realizing all its moments.

Lemma 2.1 (Elementary moment bounds). *For every array satisfying (2.1), one has $|m_J| \leq 1$ for $|J| \leq 2D$. The value under $\mathcal{E}$ of any product of at most $2D$ distinct input bits has magnitude at most one. On polynomials of degree at most D , the form*

$$\langle P, R \rangle_{\mathcal{E}} = \mathcal{E}[P \overline{R}]$$

is a positive semidefinite Hermitian form, linear in its first argument. It satisfies Cauchy–Schwarz and becomes an inner product after quotienting by its zero-norm subspace.

Proof. The empty moment equals one. Every nonempty J with $|J| \leq 2D$ can be partitioned as $J = I \sqcup K$ with $|I|, |K| \leq D$. The I, K principal submatrix in (2.1) has diagonal entries one and off-diagonal entry m_J , giving $|m_J| \leq 1$. Expanding a product of unfixed bits as $2^{-t} \prod (1 + Y_i)$ is a convex combination of these sign monomials; fixed bits either remove factors or make the product zero. This proves the second assertion. The final assertion is the Gram-form interpretation of the real symmetric positive semidefinite matrix (2.1), extended to complex coefficient vectors. Cauchy–Schwarz follows by applying positivity to $P + zR$ for $z \in \mathbb{C}$; it also shows that every zero-norm vector is orthogonal to every vector, so the quotient is well defined. $\square$

Fix a true input $x \in \{0, 1\}^n$ and a prefix u of length d with

$$u_i = x_i \quad (i < d), \quad u_d = 1 - x_d.$$

For a positive moment functional fixing u , write

$$\Delta[P] = \mathcal{E}[P] - P(x).$$

Then $\Delta[X_d] = \pm 1$, and $\Delta[P] = 0$ whenever P uses only positions strictly before d . If P is a linear combination of bit monomials of degree at most $2D$, Lemma 2.1 gives

$$|\Delta[P]| \leq 2 \sum_A |c_A| \quad \text{when } P = \sum_A c_A \prod_{i \in A} X_i. \quad (2.2)$$

For $p < 1$, introduce

$$q_* = \max(q, 1/n), \quad Q = 1 + \log(1/q_*), \quad Y = 1 + \frac{\log n}{Q}. \quad (2.3)$$

When $q \geq 1/n$, $Y = X(n, p)$. When $q < 1/n$, $1 \leq X(n, p) \leq Y < 2$. Hence

$$X(n, p) \leq Y \leq 2X(n, p), \quad QY = Q + \log n \leq C \log n$$

for an absolute constant C , since $n \geq 2$.

Theorem 2.2 (Separation from a wrong prefix). *There exist absolute constants $C_{\text{sep}} \geq 1$ and an integer $H \geq 1$ with the following property. For $n \geq 2$ and $0 < p < 1$, define*

$$D_* = 4^{\lceil \log_2 Y \rceil + H} \leq 4^{H+1} Y^2,$$

using (2.3). For every $x \in \{0, 1\}^n$, every $d \in [n]$, and every positive moment functional of order $2D$ fixing $x_1, \dots, x_{d-1}, 1 - x_d$, where $D \geq D_$, some nonempty $I \subseteq [n]$ with $|I| \leq D_*$ satisfies*

$$|\mathcal{E}[T_I] - T_I(x)| > \exp(-C_{\text{sep}} B(n, p)).$$

The theorem concerns every feasible array, including arrays not represented by distributions. Its proof occupies the next two sections. Once it is available, empirical trace moments rule out every wrong extension of a correct prefix. Section 5 shows how to find the feasible extension with an explicit bounded computation.

2.3 Exponentially weighted tuple statistics

For the analytic proof, fix $x, d, \mathcal{E}$ as above. For $1 \leq k \leq 2D$, an *order- k feature* is a sequence of polynomials

$$f_j = \sum_{i_1 < \dots < i_k = j} X_{i_1} \cdots X_{i_k} \prod_{h=1}^{k-1} z_h^{i_{h+1} - i_h}, \quad j \in [n], \quad (2.4)$$

where $z_h = \exp(-a_h + i\theta_h)$, $a_h > 0$, and $\theta_h \in \mathbb{R}$. For $k = 1$ this means $f_j = X_j$. Put

$$H_f = \prod_{h=1}^{k-1} (1 + 1/a_h).$$

Summing geometric series in the positive gaps shows that $H_f \geq 1$ bounds the sum of the absolute coefficients of each f_j , since $(1 - e^{-a})^{-1} \leq 1 + 1/a$ for $a > 0$. For $s > 0$ and $\omega \in \mathbb{R}$, define its signal by

$$\mathcal{S}_f(s, \omega) = \sum_{j=1}^n \Delta[f_j] e^{(-s+i\omega)(j-d)}. \quad (2.5)$$

Every monomial in f_j uses positions at most j , so $\Delta[f_j] = 0$ for $j < d$. Thus, despite the centered exponent, only nonnegative distances contribute to (2.5). The purpose of the multiscale argument is to preserve a detectable signal while reducing s and $|\omega|$, at the cost of increasing k .

3 Moving a signal to a larger scale

The signal in (2.5) discounts positions beyond distance about $1/s$ from the first disagreement. The next proposition moves such a signal to a much smaller decay parameter. It permits an increase in feature order, but keeps the loss in the logarithm of the signal below a factor of eight. Controlling the angles of the new parameters is equally important: those bounds will later permit passage from input coordinates to trace coordinates. Enlarging spatial scales while increasing statistic order follows the strategy of Burudgunte, Valiant, and Wang [4]. Here the comparison functional may be a positive truncated moment functional, and the scale step uses Gram matrices of polynomial vectors.

Proposition 3.1 (One multiscale step). *There are absolute constants $0 < \beta, \eta_0 < 1$ and $C_0, C_1 \geq 1$ with the following property. Let f be an order- k feature from (2.4), and let Δ arise from a moment functional of order $2D$ with the prefix conditions of Section 2. Suppose*

$$0 < S \leq 4, \quad 0 < S' \leq \min\{1, \eta_0 S\}, \quad 4k \leq D,$$

and, for some $\beta S \leq s \leq S$ and $|\omega| \leq 2.002s$,

$$|\mathcal{S}_f(s, \omega)| \geq e^{-E}, \quad E \geq C_0 \left(1 + k \log(2k) + \log H_f + \log \frac{1}{S'} \right). \quad (3.1)$$

Then an order- k' feature f' , with $1 \leq k' \leq 4k$, satisfies

$$|\mathcal{S}_{f'}(s', \omega')| \geq e^{-7.98E}, \quad \beta S' \leq s' \leq S', \quad |\omega'| \leq 2.002s', \quad (3.2)$$

and

$$H_{f'} \leq H_f^4 \left(1 + \frac{1}{\beta S'} \right)^3. \quad (3.3)$$

Moreover, there are $1 \leq m \leq 4$ lists, each equal to the old parameter list $(z_1, \dots, z_{k-1})$ or its complex conjugate, and at most $m - 1$ added parameters, with the following property. Each new inner-gap parameter is a nonempty product using at most one entry from each list and at most one added parameter. Every added parameter $e^{-a+i\theta}$ obeys

$$a \geq \beta S', \quad |\theta| \leq C_1 S. \quad (3.4)$$

The proof has two parts. First, a broad Gaussian window turns the original signal into Fourier sums. Positivity of the moment functional then forces a product of at most four such sums to retain a signal while its total frequency is small. Second, expanding that product and deforming Gaussian contours produces a single feature with the asserted parameters. The first part occupies the present subsections; the conversion is carried out in Section 3.4.

3.1 A Gaussian window and its logarithmic bound

Throughout the proof of Proposition 3.1, set

$$\sigma = \frac{K\sqrt{E}}{S'}, \quad W(l) = \exp\left(-\frac{l^2}{2\sigma^2}\right), \quad b = \beta S', \quad (3.5)$$

where K is a sufficiently large absolute constant. The constants will be chosen in the order K , then β sufficiently small, then η_0 sufficiently small, and finally C_0 sufficiently large. The choices of K and β will also enforce the contour estimates below.

We use the following consequence of (3.1) repeatedly. The logarithm of any fixed product of powers of $H_f, 1 + \sigma, 1/S', 1/b$, together with a factor $\exp(O(k \log(2k)))$, is bounded by

$$C \left(1 + k \log(2k) + \log H_f + \log \frac{1}{S'}\right) + C \log E,$$

where C depends only on constants already chosen. Increasing C_0 makes this at most any prescribed positive multiple of E . Only finitely many such estimates will be required, so a single final choice of C_0 suffices.

For real t , define the polynomial and its evaluation

$$A(t) = \sum_{j=1}^n f_j W(j-d) e^{it(j-d)}, \quad B_x(t) = A(t)(x). \quad (3.6)$$

The coefficient bound for f_j and the Gaussian sum imply that

$$|B_x(t)|, \quad |\mathcal{E}[A(t)]|, \quad |\Delta[A(t)]| \leq U := CH_f(1 + \sigma), \quad (3.7)$$

with C absolute and chosen so that $U \geq 1$. By prefix cancellation,

$$P(z) := \sum_{j=d}^n \Delta[f_j] W(j-d) z^{j-d} \quad \text{satisfies} \quad P(e^{it}) = \Delta[A(t)].$$

We first show that P retains a signal at almost the original decay:

$$|P(e^{-s_1+i\omega})| \geq \frac{1}{2} e^{-E} \quad \text{for some} \quad |s_1/s - 1| \leq 10^{-4}. \quad (3.8)$$

Let V be a centered real normal variable with variance σ^{-2} . Since $W(l)\mathbf{E}e^{Vl} = 1$,

$$\mathcal{S}_f(s, \omega) = \mathbf{E} P(e^{-s+V+i\omega}).$$

Write $\delta = 10^{-4}$. Exponential tilting gives

$$W(l)\mathbf{E}[e^{Vl}\mathbf{1}_{\{|V|>\delta s\}}] = \Pr(|N(l/\sigma^2, \sigma^{-2})| > \delta s).$$

For $0 \leq l \leq \delta s \sigma^2/2$, this probability is at most $2e^{-c\delta^2 s^2 \sigma^2}$. For larger l , use the bound one and sum e^{-sl} . Because $|\Delta[f_j]| \leq 2H_f$, the contribution from $|V| > \delta s$ is at most

$$2H_f(1 + 1/s) \left(2e^{-c\delta^2 s^2 \sigma^2} + e^{-\delta s^2 \sigma^2/2} \right). \quad (3.9)$$

Here $1/s \leq 1/(\beta S')$, and $s^2 \sigma^2 \geq \beta^2 K^2 E/\eta_0^2$. Choose η_0 sufficiently small and then C_0 sufficiently large to make (3.9) less than e^{-2E} . The integral over $|V| \leq \delta s$ consequently has magnitude at least $e^{-E} - e^{-2E} \geq e^{-E}/2$. Its maximum integrand magnitude is at least that large, proving (3.8).

Set

$$\Lambda(t) = \log \frac{U}{|P(e^{it})|} \geq 0,$$

with value $+\infty$ at a boundary zero. We claim

$$\int_{\mathbb{R}} \Lambda(t) \frac{s_1 dt}{\pi(s_1^2 + (t - \omega)^2)} \leq 1.001E. \quad (3.10)$$

For completeness, the Cauchy Fourier transform is

$$\int_{\mathbb{R}} \frac{e^{i\xi v}}{\pi(1 + v^2)} dv = e^{-|\xi|}. \quad (3.11)$$

For $\xi > 0$, close the contour in the upper half-plane and take the residue at i ; for $\xi < 0$ use the lower half-plane, and for $\xi = 0$ integrate directly. Thus periodizing the density in (3.10) gives the circle Poisson measure at $e^{-s_1 + i\omega}$.

To apply it, factor the nonzero polynomial P into linear factors. A factor with a root outside the disk has harmonic logarithmic magnitude inside. For a root a inside, the boundary identity $|e^{it} - a| = |1 - \bar{a}e^{it}|$ and the inequality

$$|1 - \bar{a}z|^2 - |z - a|^2 = (1 - |a|^2)(1 - |z|^2) \geq 0$$

give the corresponding upper bound by its Poisson integral. Roots on the circle follow by a radial limit; their logarithmic singularities are integrable. Summing over factors yields

$$\int_{\mathbb{R}} \Lambda(t) \frac{s_1 dt}{\pi(s_1^2 + (t - \omega)^2)} \leq \log U - \log |P(e^{-s_1 + i\omega})| \leq E + \log(2U).$$

The last term is at most $1.001E$ by the choice of C_0 , proving the claim.

3.2 Frequencies at prescribed differences

We seek a frequency $\lambda \in [-2s_1, 2s_1]$ with a large signal and pairs of frequencies with every prescribed difference in that interval. The product $A(t)\overline{A(t')}$ has signed frequency $t - t'$. After normalization by their evaluations at x , these products will supply the interior vectors of a chain from 1 to a normalized copy of $A(\lambda)$, with consecutive frequency labels differing by at most b . Each neighboring inner product will then have signed frequency of magnitude at most b , even though $|\lambda|$ may be much larger. The logarithmic estimate makes the required existence of pairs a question about pairing Cauchy mass.

Lemma 3.2 (Cauchy matching). *Let $w(v) = 1/(\pi(1+v^2))$. For every $r \geq 0$ there is a finite positive measure μ_r on $\mathbb{R}^2$, supported on $v - v' = r$, such that the sum of its two marginals is bounded above by $w(v) dv$. For $r > 0$ its total mass is*

$$2\mu_r(\mathbb{R}^2) = 1 - \frac{4}{\pi} \arctan(e^{-\pi/r}); \quad (3.12)$$

for $r = 0$ its total mass is $1/2$. In particular, the quantity in (3.12) is greater than 0.737 when $0 < r \leq 2$.

Proof. For $r = 0$, put half the measure $w(v) dv$ on the diagonal. Suppose $r > 0$, and partition the line into chains $v + jr$, $j \in \mathbb{Z}$, with $-r/2 \leq v < r/2$. Abbreviate $w_j = w(v + jr)$. On the edge from j to $j - 1$, for $j \geq 1$, put weight

$$e_j = \sum_{m \geq 0} (-1)^m w_{j+m}.$$

These weights are nonnegative, since the summands decrease along the tail. Moreover $e_j + e_{j+1} = w_j$, so the edges exhaust the weight at every positive vertex. Construct the negative-side edges in the same way, proceeding from the negative tail toward zero. The only possibly unused weight is at the central vertex, where its value is

$$R(v) = \sum_{j \in \mathbb{Z}} (-1)^j w(v + jr).$$

To check that no weight has been overused, periodize w at period $2r$ and subtract its translate by r . By (3.11), with $\tau = e^{-\pi/r}$, this gives the absolutely convergent Fourier series

$$R(v) = \frac{2}{r} \sum_{m \geq 0} \tau^{2m+1} \cos \frac{(2m+1)\pi v}{r}.$$

Writing $a = \pi v/r$, the sum equals

$$\operatorname{Re} \frac{\tau e^{ia}}{1 - \tau^2 e^{2ia}} = \frac{\tau(1 - \tau^2) \cos a}{1 - 2\tau^2 \cos(2a) + \tau^4} \geq 0,$$

since $|a| \leq \pi/2$. Orient every edge with its larger endpoint first and integrate the edge weights with respect to dv over $[-r/2, r/2)$. The resulting measure has the asserted marginal domination. Its uncovered mass is

$$\int_{-r/2}^{r/2} R(v) dv = \frac{4}{\pi} \sum_{m \geq 0} \frac{(-1)^m \tau^{2m+1}}{2m+1} = \frac{4}{\pi} \arctan \tau,$$

which proves (3.12). The expression decreases with r , and at $r = 2$ it exceeds 0.737. For example, $e^{-\pi/2} < 0.209$ together with $\arctan u \leq u - u^3/3 + u^5/5$ for $0 \leq u \leq 1$ proves this numerical bound. $\square$

Lemma 3.3 (Frequencies from the logarithmic bound). *Let $s_1, E > 0$, let $|\omega| \leq 2.003s_1$, and let $\Lambda : \mathbb{R} \rightarrow [0, \infty]$ be measurable and satisfy (3.10). There is a real λ with*

$$|\lambda| \leq 2s_1, \quad \Lambda(\lambda) \leq 2.40E. \quad (3.13)$$

There is also an absolute C_2 such that, for every $|y| \leq 2s_1$, some real t, t' satisfy

$$t - t' = y, \quad |t|, |t'| \leq C_2 s_1, \quad \Lambda(t) + \Lambda(t') \leq 2.74E. \quad (3.14)$$

Proof. The Cauchy mass of $[-2s_1, 2s_1]$ is at least

$$\frac{\arctan(2 - 2.003) + \arctan(2 + 2.003)}{\pi} > 0.418.$$

Because $1.001/0.418 < 2.40$, averaging proves (3.13).

For (3.14), use Lemma 3.2 with $r = |y|/s_1 \leq 2$ and then substitute $t = \omega + s_1 v$. Reverse the orientation if needed. The sum of the two marginals is bounded by the Cauchy measure in (3.10), while twice the total pair mass is greater than 0.737. Discard pairs with an endpoint outside $[-C_2 s_1, C_2 s_1]$. Twice the discarded pair mass is at most twice the Cauchy tail outside that interval, which tends uniformly to zero as $C_2 \rightarrow \infty$, since $|\omega|/s_1 \leq 2.003$. Choose an absolute C_2 so the retained pair mass exceeds $0.733/2$. Its integral of $\Lambda(t) + \Lambda(t')$ is at most $1.001E$. Some retained pair consequently has sum at most

$$\frac{2(1.001)}{0.733}E < 2.74E,$$

as required. The diagonal construction gives the same argument when $y = 0$. $\square$

3.3 A product with small total frequency

The window parameters satisfy $|\omega|/s_1 \leq 2.002/0.9999 < 2.003$, so Lemma 3.3 applies to our Λ . Individual frequencies in (3.14) need not be small. Their differences, however, can be prescribed. We use these pairs to build a chain of polynomial vectors whose neighboring inner products have small total frequency. The following Hilbert-space fact explains why such a chain preserves the needed exponential accuracy.

Lemma 3.4 (Gram chaining). *Let $v_0, \dots, v_J$, with $J \geq 1$, lie in a complex Hilbert space whose inner product is linear in its first argument. Suppose $\|v_0\| = 1$ and, for some $\xi \geq 0$,*

$$\|v_i\|^2 - 1 \leq \xi \quad (0 \leq i \leq J), \quad |\langle v_i, v_{i-1} \rangle - 1| \leq \xi \quad (1 \leq i \leq J).$$

Then

$$|\langle v_J, v_0 \rangle - 1| \leq 2J^2\xi.$$

Proof. Put $h_i = v_i - v_{i-1}$. The hypotheses give

$$\|h_i\| \leq 2\sqrt{\xi}, \quad |\langle h_i, v_{i-1} \rangle| \leq 2\xi, \quad \|v_{i-1} - v_0\| \leq 2(i-1)\sqrt{\xi}.$$

Thus Cauchy–Schwarz implies

$$|\langle h_i, v_0 \rangle| \leq 2\xi + 4(i-1)\xi.$$

Sum over i and use $\langle v_0, v_0 \rangle = 1$. $\square$

Choose an absolute C_3 large enough that all frequencies supplied by Lemma 3.3 have magnitude at most $C_3 S$. A product of one to four factors, each of the form $A(t)$ or $\overline{A(t)}$ with $|t| \leq C_3 S$, is called *balanced* if the sum of its signed frequencies has magnitude at most b ; conjugation changes the sign of a frequency.

Lemma 3.5 (A balanced product retains a signal). *Some balanced product G satisfies*

$$|\Delta[G]| \geq e^{-7.94E}. \tag{3.15}$$

Proof. Suppose to the contrary that every balanced product has signal less than $\epsilon = e^{-7.94E}$. The squared modulus $|A(t)|^2$ is one such product. Positivity of $\mathcal{E}$, applied to $A(t)$ and 1, gives

$$|\mathcal{E}[A(t)]|^2 \leq \mathcal{E}[|A(t)|^2] \leq |B_x(t)|^2 + \epsilon,$$

and hence

$$|\Delta[A(t)]| \leq 2|B_x(t)| + \sqrt{\epsilon}. \quad (3.16)$$

At λ from (3.13), this implies

$$|B_x(\lambda)| \geq e^{-2.41E}. \quad (3.17)$$

Indeed $|\Delta[A(\lambda)]| \geq e^{-2.40E}$, whereas $\sqrt{\epsilon} = e^{-3.97E}$; fixed numerical factors are absorbed by the absolute lower bound on E in (3.1). For a pair in (3.14), nonnegativity of Λ implies that both corresponding signals are at least $e^{-2.74E}$, and

$$|\Delta[A(t)]\Delta[A(t')]| = U^2 e^{-\Lambda(t)-\Lambda(t')} \geq e^{-2.74E}.$$

Applying (3.16) to each therefore gives

$$|B_x(t)\overline{B_x(t')}| \geq e^{-2.75E}. \quad (3.18)$$

For example, each evaluation is at least one third of its corresponding signal once E is large; the resulting factor $1/9$ is absorbed in $e^{-0.01E}$.

Take a monotone chain $0 = y_0, y_1, \dots, y_J = \lambda$ with consecutive distances at most b and

$$1 \leq J \leq C(1 + S/b).$$

If $\lambda = 0$, use just two endpoints and $J = 1$. For every interior y_i , choose a pair from (3.14) and define polynomials

$$V_0 = 1, \quad V_i = \frac{A(t)\overline{A(t')}}{B_x(t)\overline{B_x(t')}} \quad (0 < i < J), \quad V_J = \frac{A(\lambda)}{B_x(\lambda)}.$$

The denominators are nonzero by (3.17)–(3.18), and every $V_i(x)$ equals one. Regard these polynomials as vectors v_i in the Hilbert space induced by $\mathcal{E}$. They have degree at most $2k \leq D$.

For $1 \leq i \leq J$, the norm square of v_i and its inner product with v_{i-1} are expectations of normalized balanced products: their signed frequencies are respectively zero and $y_i - y_{i-1}$, and each product has at most four factors. The magnitude of its normalizing denominator is at least $e^{-5.50E}$. Our contradictory assumption therefore gives

$$|||v_i||^2 - 1| \leq \xi, \quad |\langle v_i, v_{i-1} \rangle - 1| \leq \xi, \quad \xi = \epsilon e^{5.50E} = e^{-2.44E}.$$

Since $v_0 = 1$ has norm one, Lemma 3.4 yields

$$|\Delta[A(\lambda)]| = |B_x(\lambda)| |\langle v_J, v_0 \rangle - 1| \leq 2UJ^2 e^{-2.44E} < e^{-2.40E}.$$

The final strict inequality follows by increasing C_0 , because $\log(2UJ^2)$ is less than $0.04E$ by (3.1). This contradicts (3.13) and proves the lemma. $\square$

We have obtained a product of at most four windowed feature sums with signal at least $e^{-7.94E}$ and total frequency at most b . It remains to turn it into one feature while keeping both its decay and frequency of order S' . The Gaussian contour construction below supplies that final step of Proposition 3.1.

3.4 Converting a balanced product to a feature

Lemma 3.5 supplies a product of at most four windowed Fourier sums with a substantial discrepancy and small total frequency. We now express that product as an integral of features. The contour will place their outer parameters in the sector required by Proposition 3.1; the remaining issue will be to control the damping factors on their inner gaps.

Fix a product supplied by Lemma 3.5, and let $1 \leq m \leq 4$ be its number of factors. Expand each factor in the tuples defining f . An *interleaving* specifies the ordering, including all ties, of the mk labeled positions in these tuples. There are at most $(mk)^{mk} = \exp(O(k \log(2k)))$ interleavings: assigning to each labeled position its rank among the distinct positions gives such a bound. Only interleavings consistent with strict increase inside each old tuple are retained. Fix one interleaving, and relabel its copies so that their endpoints satisfy $j_1 \leq \dots \leq j_m$. Equal endpoints may be ordered arbitrarily. After absorbing conjugation into the signs of the frequencies, their window weight is

$$\prod_{r=1}^m W(j_r - d) \exp(it_r(j_r - d)), \quad |t_r| \leq C_3 S, \quad \left| \sum_{r=1}^m t_r \right| \leq b. \quad (3.19)$$

Conjugation also conjugates the old inner-gap parameters, leaving their dampings unchanged.

A Gaussian contour with controlled total frequency. Recall that $\sigma = K\sqrt{E}/S'$ and $b = \beta S'$. For every real l ,

$$W(l) = \frac{\sigma}{i\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(\sigma^2 v^2/2 + vl) dv. \quad (3.20)$$

Indeed, writing $v = iu$ gives the Gaussian Fourier transform. Its value at $l = 0$ is one, and integration by parts shows that its derivative in l is $-l/\sigma^2$ times itself, proving the formula.

Set $\rho = 1/2.002$. For $u = (u_1, \dots, u_m) \in \mathbb{R}^m$, define

$$\Omega = \sum_{r=1}^m (u_r + t_r), \quad v_r = b + iu_r \quad (r < m), \quad v_m = -mb - \rho|\Omega| + iu_m. \quad (3.21)$$

These real parts have partial sums rb for $r < m$ and total sum $-b - \rho|\Omega|$. When the endpoint exponent is centered at j_m , the total sum gives the real part of its outer coefficient, while the negatives of the partial sums give the real parts of the coefficients on consecutive endpoint gaps. The contour is therefore designed to give outer decay $b + \rho|\Omega|$ and positive inner dampings rb ; we verify the exact rewriting below.

For each fixed tuple of positions, first shift the first $m - 1$ contours in (3.20) to their fixed vertical lines $\operatorname{Re} v_r = b$, leaving the last contour on the imaginary axis. The connecting horizontal integrals vanish by Gaussian decay. Next fix $(u_1, \dots, u_{m-1})$ and deform the last contour to the piecewise linear graph in (3.21). Its slopes have absolute value $\rho < 1$. Along horizontal segments joining its truncations to the imaginary axis at height $\pm T$, the Gaussian exponent has real part at most $-cT^2 + O(T)$ for some $c > 0$. The factor $\exp(v_m(j_m - d))$ contributes only a linear term in T , so these connecting integrals also vanish. The integrands are entire, and hence no residues occur. This proves the identities as iterated integrals; absolute joint convergence will follow from the next bound.

Let ν be the resulting complex measure on $\mathbb{R}^m$ after the factors depending on positions have been removed. More explicitly, outside the measure-zero set $\Omega = 0$,

$$d\nu(u) = \left(\frac{\sigma}{\sqrt{2\pi}} \right)^m \exp \left(\frac{\sigma^2}{2} \sum_{r=1}^m v_r^2 \right) (1 + i\rho \operatorname{sgn} \Omega) du. \quad (3.22)$$

The last factor is the differential of the bent contour divided by $i du_m$. We next obtain estimates uniform over the frequencies in (3.19). These estimates also establish absolute convergence of the joint integrals.

Put $\delta_0 = 1 - 4\rho^2 > 0$ and $c_1 = \delta_0/2$. Since $m \leq 4$ and $|\sum t_r| \leq b$, Young's inequality gives an absolute C_4 such that

$$\begin{aligned} \sum_{r=1}^m (\operatorname{Re} v_r)^2 &= (m-1)b^2 + (mb + \rho|\Omega|)^2 \\ &\leq 3b^2 + \left(2\rho \sqrt{\sum_r u_r^2} + (4+\rho)b \right)^2 \\ &\leq (1-c_1) \sum_r u_r^2 + C_4 b^2. \end{aligned} \quad (3.23)$$

The strict inequality $4\rho^2 < 1$ is essential here: summing at most four imaginary coordinates can multiply their squared norm by four, and the sector constant 2.002 leaves a positive Gaussian decay margin. For example, $C_4 = 3 + (4+\rho)^2(1+8\rho^2/\delta_0)$ suffices. Together with (3.22), this proves

$$|d\nu(u)| \leq C\sigma^m \exp\left(\frac{C_4\sigma^2 b^2}{2} - \frac{c_1\sigma^2}{2} \sum_r u_r^2\right) du.$$

For any fixed tuple of positions, the absolute value of its remaining exponential factor is at most $\exp(C_n(1 + \sum_r |u_r|))$ for a finite constant C_n . This factor is integrable against the Gaussian bound. Thus the iterated deformation also gives a jointly absolutely convergent integral, and finite tuple sums can be interchanged with it.

Write $\mathcal{U} = \{u : \max_r |u_r| \leq S'/16\}$. Integrating the last bound, and splitting its negative quadratic term into two equal parts on $\mathcal{U}^c$, gives absolute constants C, C' with

$$\begin{aligned} \|\nu\|_{\text{abs}} &\leq C \exp(C_4 K^2 \beta^2 E/2), \\ \int_{\mathcal{U}^c} |d\nu| &\leq C' \exp\left[\left(\frac{C_4 K^2 \beta^2}{2} - \frac{c_1 K^2}{1024}\right) E\right]. \end{aligned}$$

Here $\|\nu\|_{\text{abs}}$ denotes total variation; the powers σ^m cancel under Gaussian rescaling. Choose K large enough that $c_1 K^2/1024 \geq 30$, then choose $\beta \leq 1/4$ small enough that $C_4 K^2 \beta^2/2 \leq 0.001$. Increasing C_0 to absorb the fixed prefactors yields

$$\|\nu\|_{\text{abs}} \leq e^{0.005E}, \quad \int_{\mathcal{U}^c} |d\nu| \leq e^{-20E}. \quad (3.24)$$

This choice is consistent with the order of constants specified earlier: K is fixed first, β second, and η_0 may then be reduced to meet the window estimate. Only afterward is C_0 enlarged.

Outer parameters and endpoint gaps. For $c_r = v_r + it_r$, telescoping the endpoint differences gives

$$\sum_{r=1}^m c_r(j_r - d) = \left(\sum_{r=1}^m c_r\right)(j_m - d) - \sum_{r=1}^{m-1} \left(\sum_{l=1}^r c_l\right)(j_{r+1} - j_r). \quad (3.25)$$

Consequently, the largest endpoint has coefficient $-s' + i\Omega$, where

$$s' = b + \rho|\Omega|. \quad (3.26)$$

The link from j_r to j_{r+1} receives the parameter

$$\exp\left(-rb - i \sum_{l=1}^r (u_l + t_l)\right), \quad 1 \leq r < m. \quad (3.27)$$

Thus every added damping is at least b , for every u . On $\mathcal{U}$,

$$|\Omega| \leq (\beta + 1/4)S', \quad \beta S' \leq s' \leq [\beta + \rho(\beta + 1/4)]S' \leq S', \quad |\Omega| \leq 2.002s'.$$

The angle in (3.27) has absolute value at most $3(C_3 S + S'/16) \leq C_1 S$ for a fixed absolute C_1 . The outer parameters therefore satisfy the parameter inequalities in (3.2), and the endpoint links satisfy (3.4). The choice of C_1 uses only C_3 and $S' \leq S$ and imposes no further condition on K or β .

3.5 Interleavings and the cost of the inner gaps

We have obtained outer weights at the required scale. To finish, we must show that each interleaving has exactly the form of a feature, with controlled coefficient sum. One old inner edge can cross several gaps of the merged tuple, so charging its damping cost at every crossing would overcount it. A distinct-edge assignment will prevent this loss.

For our fixed interleaving, write its distinct positions as $i_1 < \dots < i_v$, where $1 \leq v \leq mk$. Form a multigraph on these vertices: include the inner edges of each old tuple and the links between consecutive endpoints in (3.27). Its edges carry their corresponding exponential parameters. It is connected, because each old tuple is a path to its endpoint, and the endpoint links join those paths. Delete loops coming from equal endpoints; their weights are one.

For each merged gap $i_{a+1} - i_a$, multiply the parameters of all edges crossing it. Each edge length is the sum of the merged gaps it crosses, so these new parameters reproduce the original weight exactly. Every merged gap is crossed by some edge, by connectivity, and hence its damping is positive. Within a single old tuple, the inner-edge intervals have disjoint interiors. The same holds for the consecutive endpoint links. It follows that each new parameter uses at most one old parameter from each copy and at most one added endpoint parameter. This is precisely the factor-count assertion in Proposition 3.1.

Ties merge repeated bit variables by the Boolean identity $X_i^r = X_i$ for $r \geq 1$. Once the interleaving is fixed, choosing arbitrary strictly increasing values for its v distinct positions specifies exactly one assignment to its labeled tuples. Thus no further restriction remains on the new tuple. Its largest position is $i_v = j_m$. Summing all these assignments gives an order- v feature, denoted $f^{\mathcal{I},u}$, with outer weight $\exp((-s' + i\Omega)(i_v - d))$.

The following elementary fact controls its height.

Lemma 3.6 (Assigning distinct edges to consecutive gaps). *Let a connected multigraph have ordered vertices $i_1 < \dots < i_v$. There is an assignment of a distinct edge to each gap (i_a, i_{a+1}) such that its assigned edge crosses that gap.*

Proof. There is nothing to prove for $v = 1$. Otherwise retain a spanning tree and orient each of its edges from its smaller to its larger endpoint. Let M be the $(v-1) \times (v-1)$ matrix whose (e, a) entry is one if tree edge e crosses gap a , and zero otherwise. This matrix is invertible. To see this, for an arbitrary vector g in its kernel set $y_1 = 0$ and $y_{a+1} = y_a + g_a$. The equation $Mg = 0$ says that y has equal values at the endpoints of every tree edge. Connectivity makes all its values zero, so $g = 0$. A nonzero term in the determinant expansion of M pairs its rows and columns through entries equal to one. That pairing is the required assignment. $\square$

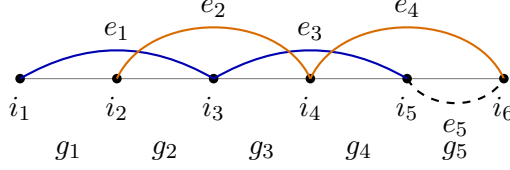


Figure 1: A schematic interleaving of the two old tuples (i_1, i_3, i_5) and (i_2, i_4, i_6) , joined by the dashed endpoint link e_5 . Here g_a denotes the gap (i_a, i_{a+1}) . The assignment $g_a \mapsto e_a$ for $1 \leq a \leq 5$ uses a distinct crossing edge for every gap. For example, e_1 crosses both g_1 and g_2 , but it is charged only to g_1 ; e_2 supplies the bound for g_2 .

Figure 1 shows why distinctness matters. Although one edge may contribute to several new parameters, it need pay for only one of their upper bounds.

Give each edge occurrence e its damping $a_e > 0$. The damping of a merged gap is the sum of the a_e over all edges crossing it, and is therefore at least that of its assigned edge. Since $1 + 1/a$ decreases with a , Lemma 3.6 implies

$$\begin{aligned} H_{f^{\mathcal{I},u}} &\leq \prod_{e \text{ assigned}} (1 + 1/a_e) \leq \prod_{e \text{ present}} (1 + 1/a_e) \\ &\leq H_f^m (1 + 1/b)^{m-1} \leq H_f^4 (1 + 1/b)^3. \end{aligned} \quad (3.28)$$

The second inequality uses that all factors are at least one; the endpoint-link dampings are $rb \geq b$. This argument includes all overlaps between old tuples and also the case of a single distinct position. In that case the height is the empty product one.

Completing the multiscale estimate. Apply Δ to the expanded product and use the integral representation for each interleaving. With $\nu_{\mathcal{I}}$ denoting the corresponding measure, we have the exact identity

$$\Delta[\text{product}] = \sum_{\mathcal{I}} \int_{\mathbb{R}^m} \mathcal{S}_{f^{\mathcal{I},u}}(s', \Omega) d\nu_{\mathcal{I}}(u). \quad (3.29)$$

All estimates above are uniform over the interleavings. The integrands have a uniform bound even outside $\mathcal{U}$. Indeed, for each fixed endpoint j , the coefficients of $f_j^{\mathcal{I},u}$ have absolute sum at most the height in (3.28). The terms with endpoint $j < d$ vanish under Δ by prefix agreement. Using $s' \geq b$ gives

$$|\mathcal{S}_{f^{\mathcal{I},u}}(s', \Omega)| \leq 2H_f^4 (1 + 1/b)^3 \sum_{l \geq 0} e^{-bl} \leq 2H_f^4 (1 + 1/b)^4. \quad (3.30)$$

In particular no exponential factor from negative distances occurs in the tail estimate.

Let N_{int} be the number of interleavings and put $V = 2H_f^4 (1 + 1/b)^4$. Hypothesis (3.1), with a sufficiently large absolute C_0 , ensures

$$E \geq 100, \quad \log N_{\text{int}} \leq 0.01E, \quad \log V \leq 0.01E.$$

Here $\log(1 + 1/b)$ is bounded by an absolute constant plus $\log(1/S')$, because $b = \beta S'$, $S' \leq 1$, and β has already been fixed. If every signal in (3.29) with $u \in \mathcal{U}$ had magnitude below $e^{-7.98E}$, then (3.24) and (3.30) would give

$$|\Delta[\text{product}]| \leq N_{\text{int}} \left(e^{(0.005-7.98)E} + V e^{-20E} \right) \leq e^{-7.965E} + e^{-19.98E} < e^{-7.94E}.$$

This contradicts Lemma 3.5. Some interleaving and some $u \in \mathcal{U}$ therefore give a feature with signal at least $e^{-7.98E}$. Its order is at most $mk \leq 4k$; (3.26) and the bounds on $\mathcal{U}$ give the required outer parameters; (3.28) gives (3.3); and the crossing description gives the claimed parameter factors and (3.4). This completes the proof of Proposition 3.1.

4 From an input signal to separation of trace moments

We prove Theorem 2.2. Proposition 3.1 preserves a detectable input-coordinate signal while reducing its decay rate. We first choose enough scales to make that rate at most $1/n$. The resulting gap parameters need not belong to the disk on which trace moments directly control the signal. A simultaneous one-variable continuation will bridge this difference.

Throughout this section $0 < p < 1$, and q_*, Q, Y are as in (2.3). Recall from Section 2 that

$$1 \leq X(n, p) \leq Y \leq 2X(n, p), \quad QY = Q + \log n \leq C \log n. \quad (4.1)$$

4.1 A scale schedule and the surviving signal

Set

$$\mu = 7.98, \quad \alpha = 0.001, \quad h = \lceil \log_2 Y \rceil + H, \quad D_* = 4^h, \quad (4.2)$$

where the absolute integer H will be fixed below. In particular, $D_* \leq 4^{H+1}Y^2$. Fix an arbitrary positive moment functional of order $2D$, where $D \geq D_*$, with first wrong position d , and let Δ be its difference from evaluation at the true word, as in Section 2. All constants chosen below are independent of this system, n , and p .

For an absolute constant $\eta > 0$, define

$$S_0 = 4, \quad S_j = \eta q_* Y^{-\alpha} S_{j-1} \min\{1, D_* S_{j-1}\} \quad (1 \leq j \leq h). \quad (4.3)$$

The linear regime of this recurrence brings $D_* S_j$ below one; its quadratic regime then makes S_j small rapidly. Both regimes will also be used to control the arguments of the final gap parameters.

Choose an integer J_* with $\alpha J_* \geq 3$, and set $H = J_* + 3$. To verify the decrease precisely, write

$$\vartheta = \eta q_* Y^{-\alpha}, \quad t_j = D_* S_j.$$

Then $t_j = \vartheta t_{j-1} \min\{1, t_{j-1}\}$, so, regardless of when the quadratic regime begins,

$$t_{J_*} \leq \vartheta^{J_*} t_0 \leq 4^{H+2} \eta^{J_*} Y^{2-\alpha J_*} \leq 1$$

once η is small enough. Require also

$$\eta \leq \min\{\eta_0, 1/4, e^{-1}\},$$

where η_0 is from Proposition 3.1. Now $\log(1/\vartheta) \geq Q$, and $t_j \leq 1$ for $j \geq J_*$. Iterating $t_j = \vartheta t_{j-1}^2$ in this range gives

$$t_h \leq \vartheta^{2^{h-J_*}-1} \leq \exp\{-Q(2^{h-J_*} - 1)\} \leq n^{-1}, \quad S_h \leq n^{-1}. \quad (4.4)$$

The last inequality follows from $2^{h-J_*} \geq 8Y$ and $Q(Y-1) = \log n$. At every transition we also have $0 < S_j \leq \min\{1, \eta_0 S_{j-1}\}$, as required by the proposition.

We need an upper bound on the cost of these small scales as well. Put $A_0 = Q + \log Y$. The recurrence implies, with $\ell_j = \max\{0, \log(1/S_j)\}$,

$$\ell_j \leq 2\ell_{j-1} + \log(1/\vartheta), \quad \ell_0 = 0, \quad \log(1/\vartheta) \leq CA_0.$$

Consequently

$$\log(1/S_j) \leq C2^j A_0 \quad (0 \leq j \leq h). \quad (4.5)$$

We can now iterate Proposition 3.1. Begin with the order-one feature $f_i^{(0)} = X_i$, at decay $s_0 = 4$ and frequency zero. The prefix agreement gives $\Delta[X_i] = 0$ for $i < d$, whereas $|\Delta[X_d]| = 1$. Since $|\Delta[X_i]| \leq 2$ for every i ,

$$|\mathcal{S}_{f^{(0)}}(4, 0)| \geq 1 - 2 \sum_{r \geq 1} e^{-4r} = 1 - \frac{2}{e^4 - 1} > \frac{1}{2}. \quad (4.6)$$

Use error exponents

$$E_j = C_5 A_0 \mu^j \quad (0 \leq j \leq h), \quad (4.7)$$

where C_5 is an absolute constant chosen after H and η .

Here are all the induction bounds. At stage j there is a feature $f^{(j)}$ of order $k_j \leq 4^j$, with

$$\begin{aligned} \beta S_j \leq s_j \leq S_j, \quad |\omega_j| \leq 2.002 s_j, \quad |\mathcal{S}_{f^{(j)}}(s_j, \omega_j)| \geq e^{-E_j}, \\ \log H_{f^{(j)}} \leq C4^j A_0. \end{aligned} \quad (4.8)$$

The initial case follows from (4.6) and $H_{f^{(0)}} = 1$, on increasing C_5 . At the transition from $j-1$ to j , the degree requirement is

$$4k_{j-1} \leq 4^j \leq D_* \leq D.$$

The feature-height conclusion of the proposition and (4.5) give

$$\log H_{f^{(j)}} \leq 4 \log H_{f^{(j-1)}} + 3 \log(1 + 1/(\beta S_j)) \leq 4 \log H_{f^{(j-1)}} + C2^j A_0.$$

Summing this recurrence proves the last bound in (4.8). To check the exponent hypothesis of the proposition, its bracket at this transition is at most

$$C(1 + j4^{j-1} + 4^{j-1}A_0 + 2^j A_0).$$

Because $\mu > 4$, the sequences $j(4/\mu)^{j-1}$ and $2^j/\mu^{j-1}$ are bounded. Thus one fixed choice of C_5 makes this bracket times C_0 at most E_{j-1} for every j . All hypotheses of Proposition 3.1 hold, and its signal bound is $e^{-\mu E_{j-1}} = e^{-E_j}$.

Write $f = f^{(h)}$, $k = k_h$, $s = s_h$, and $\omega = \omega_h$. We have obtained a signal of magnitude at least e^{-E_h} with $k \leq D_*$ and $s \leq 1/n$. The next step records where its parameters lie in the unit disk; this information will control the cost of continuing the signal to the trace domain.

4.2 Damping under multiplication

Call the parameters introduced between endpoints in each application of Proposition 3.1 *elementary parameters*. Include also the final outer parameter

$$z_0 = e^{-s+i\omega}.$$

An elementary parameter $e^{-a+i\theta}$ introduced at stage j satisfies $a \geq \beta S_j$ and $|\theta| \leq C_1 S_{j-1}$. Using (4.3), choose a sufficiently small absolute $c_2 > 0$ so that all these parameters obey

$$a \geq \gamma \min\{|\theta|, D_* \theta^2\}, \quad \gamma = c_2 q_* Y^{-\alpha} \leq 1. \quad (4.9)$$

For an explicit verification, the increasing function $u \mapsto u \min\{1, D_* u\}$ changes by at most a factor $\max\{C_1, C_1^2\}$ when u is replaced by $C_1 u$. It therefore suffices to take $c_2 \leq \beta \eta / \max\{C_1, C_1^2\}$. Shrinking c_2 further covers z_0 , since $|\omega| \leq 2.002s$. Conjugating any elementary parameter preserves (4.9).

Each new inner gap contains at most one parameter from each of at most four copies of the preceding feature, and at most one new elementary parameter. Count repeated factors with multiplicity; conjugation does not change the count. Starting with no inner parameters, the maximum number N_j of elementary factors in a stage- j inner gap satisfies

$$N_0 = 0, \quad N_j \leq 4N_{j-1} + 1, \quad N_j \leq \frac{4^j - 1}{3}.$$

In particular, an inner gap of f , even after multiplication by z_0 , contains at most $(4^h + 2)/3 \leq D_*$ elementary factors.

The following elementary lemma explains why this count matters. The Cayley map $z \mapsto (1+z)/(1-z)$ sends the unit disk to the right half-plane; we need a quantitative distance from its boundary.

Lemma 4.1 (Products with controlled damping). *Let $N \geq 1$ be an integer and $0 < \gamma \leq 1$. Suppose z is a nonempty product of at most N numbers $e^{-a_j+i\theta_j}$, where $a_j > 0$, $\theta_j \in \mathbb{R}$, and*

$$a_j \geq \gamma \min\{|\theta_j|, N\theta_j^2\}.$$

There is an absolute $c_3 > 0$ such that

$$\operatorname{Re} \frac{1+z}{1-z} \geq c_3 \gamma.$$

Proof. Write $z = e^{-T+i\psi}$, where $T = \sum_j a_j > 0$ and $\psi = \sum_j \theta_j$. Let

$$U = \sum_{|\theta_j| \geq 1/N} |\theta_j|, \quad V = \sum_{|\theta_j| < 1/N} |\theta_j|.$$

Summing the hypotheses and applying Cauchy–Schwarz to at most N small angles yields

$$T \geq \gamma \left(U + N \sum_{|\theta_j| < 1/N} \theta_j^2 \right) \geq \gamma(U + V^2).$$

If $U \geq 1$ or $V \geq 1$, the last bracket is at least one. Otherwise $U + V^2 \geq U^2 + V^2 \geq (U + V)^2/2$. Since $|\psi| \leq U + V$, in all cases

$$T \geq \frac{1}{2} \gamma \min\{1, \psi^2\}.$$

For $0 < T \leq 1$ it follows that

$$1 - |z|^2 \geq cT, \quad |1 - z|^2 \leq C(T^2 + \min\{1, \psi^2\}) \leq C'T/\gamma.$$

Their ratio is $\operatorname{Re}((1+z)/(1-z))$. If $T > 1$, that ratio is at least $(1 - e^{-2})/4$, which also suffices because $\gamma \leq 1$. $\square$

4.3 The exact trace generating identity

Let $z_1, \dots, z_{k-1}$ denote the inner gap parameters of the final feature f . Define the polynomial in k complex variables

$$F(Z_0, \dots, Z_{k-1}) = \sum_{1 \leq i_1 < \dots < i_k \leq n} \Delta \left[\prod_{r=1}^k X_{i_r} \right] Z_0^{i_1} \prod_{r=1}^{k-1} Z_r^{i_{r+1} - i_r}. \quad (4.10)$$

At the point

$$Z_0^* = z_0, \quad Z_r^* = z_r z_0 \quad (1 \leq r < k),$$

the exponents of z_0 telescope to i_k , so

$$F(Z^*) = z_0^d \mathcal{S}_f(s, \omega), \quad |F(Z^*)| \geq e^{-E_h - 1}. \quad (4.11)$$

Here $sd \leq nS_h \leq 1$. Each coordinate of Z^* is a nonempty product of at most D_* elementary parameters, and hence Lemma 4.1 gives

$$\operatorname{Re} \frac{1 + Z_r^*}{1 - Z_r^*} \geq c_3 \gamma \quad (0 \leq r < k). \quad (4.12)$$

We shall also need a bound on the coordinates' collective proximity to the unit circle. If $z_r = e^{-a_r + i\theta_r}$, then

$$\frac{1}{1 - |Z_r^*|} \leq 1 + \frac{1}{a_r + s} \leq 1 + \frac{1}{a_r} \quad (r \geq 1), \quad \frac{1}{1 - |Z_0^*|} \leq 1 + \frac{1}{s}.$$

These inequalities follow from $e^u \geq 1 + u$ for $u > 0$. The height bound in (4.8) and (4.5) therefore imply

$$W_* := \log \prod_{r=0}^{k-1} \frac{1}{1 - |Z_r^*|} \leq \log H_f + \log(1 + 1/(\beta S_h)) \leq C 4^h A_0. \quad (4.13)$$

We next express the same polynomial in terms of observable trace moments on a different domain. This uses the multivariate trace generating identity of [9, Section 5.2]; see also [7, Fact 20]. We give the required counting argument directly. For a rank tuple $\ell = (\ell_1, \dots, \ell_k)$ with $1 \leq \ell_1 < \dots < \ell_k \leq n$, abbreviate $T_\ell = T_{\{\ell_1, \dots, \ell_k\}}$. For $u = (u_0, \dots, u_{k-1}) \in \mathbb{C}^k$, set

$$G(u) = \sum_{1 \leq \ell_1 < \dots < \ell_k \leq n} \Delta[T_\ell] u_0^{\ell_1 - 1} \prod_{r=1}^{k-1} u_r^{\ell_{r+1} - \ell_r - 1}.$$

Then, with $Z_r = q + pu_r$, the exact polynomial identity is

$$\left(\prod_{r=0}^{k-1} Z_r \right) G(u) = p^k F(Z). \quad (4.14)$$

To prove it, fix input positions $i_1 < \dots < i_k$ that supply the specified retained bits. Their own retention contributes p^k . Every input position before i_1 contributes $q + pu_0$, according to whether it is deleted or retained; there are $i_1 - 1$ such positions. Similarly, the $i_{r+1} - i_r - 1$ positions in the r th intervening gap each contribute $q + pu_r$. Positions after i_k contribute a total factor of one. The retained counts in these intervals are exactly $\ell_1 - 1$ and $\ell_{r+1} - \ell_r - 1$. Summing over all input tuples and applying Δ gives $G(u)$ as p^k times the polynomial in (4.10) with every displayed gap exponent reduced by one. Multiplication by $\prod_r Z_r$ proves (4.14), including points where some $Z_r = 0$.

Suppose, for the moment, that

$$|\Delta[T_I]| \leq \varepsilon \quad \text{for every } I \subseteq \{1, \dots, n\} \text{ with } 1 \leq |I| \leq D_*, \quad (4.15)$$

where $\varepsilon > 0$. Since $k \leq D_*$, the rank sum has at most n^k terms, so $|G(u)| \leq n^k \varepsilon$ when all $|u_r| \leq 1$. On this polydisk $|Z_r| \leq q + p = 1$. Thus (4.14) gives

$$|F(Z)| \leq (n/p)^k \varepsilon \quad \text{if } |Z_r - q| \leq p \quad (0 \leq r < k). \quad (4.16)$$

This calculation uses only ordinary trace ranks and the convention that a rank product is zero when the trace is too short.

4.4 Simultaneous continuation and the separation bound

The lower bound (4.11) and the upper bound (4.16) concern different points. In Cayley coordinates the disk $|Z - q| \leq p$ becomes a half-plane:

$$|Z - q| \leq p \iff \operatorname{Re} \frac{1+Z}{1-Z} \geq \frac{q}{p} \quad (|Z| < 1). \quad (4.17)$$

Indeed, writing $w = (1+Z)/(1-Z)$ and expanding $|pw - (1+q)|^2 \leq p^2|w+1|^2$ reduces it to $4q \leq 4p \operatorname{Re} w$. We will move all coordinates to this half-plane using one complex variable. A single strip interpolation avoids incurring a separate continuation loss for every coordinate.

Write

$$w_r = \frac{1+Z_r^*}{1-Z_r^*} = \xi_r + i\zeta_r, \quad \xi_r \geq c_3\gamma,$$

where $\xi_r, \zeta_r \in \mathbb{R}$. For $v \in \mathbb{C}$ set

$$w_r(v) = (1+v)\xi_r + i\zeta_r, \quad Z_r(v) = \frac{w_r(v) - 1}{w_r(v) + 1}, \quad R = 1 + \frac{q}{pc_3\gamma}. \quad (4.18)$$

Figure 2 shows how a common real value of v moves two possible starting coordinates into the trace half-plane. The proof uses the analytic maps (4.18) for complex v throughout a strip.

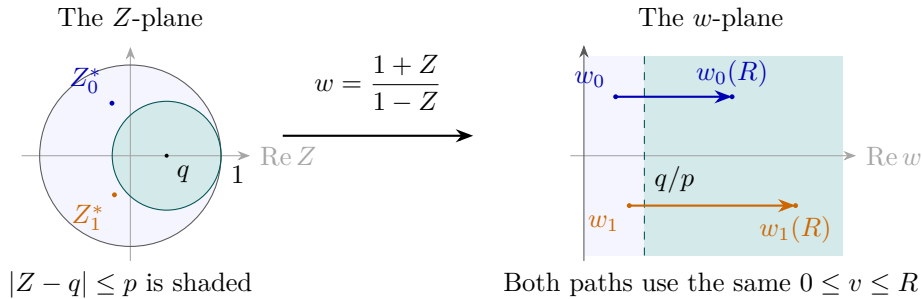


Figure 2: Schematic geometry of the simultaneous continuation. The Cayley map turns the deletion disk into $\operatorname{Re} w \geq q/p$. Two possible starting coordinates are shown. The horizontal arrows have different lengths but share the same parameter: $w_r(v) = (1+v)\operatorname{Re} w_r + i\operatorname{Im} w_r$. Only their real traces $0 \leq v \leq R$ are drawn. The proof interpolates $F(Z_0(v), \dots, Z_{k-1}(v))$ on a single complex strip, simultaneously for every coordinate.

On the closed strip $-1/2 \leq \operatorname{Re} v \leq R$ we have $\operatorname{Re} w_r(v) = (1 + \operatorname{Re} v)\xi_r > 0$, so all $Z_r(v)$ lie in the unit disk and depend analytically on v . At $v = 0$ they equal Z_r^* . On the right boundary,

$$\operatorname{Re} w_r(v) = (1 + R)\xi_r \geq q/p,$$

and therefore (4.16) applies. Moreover

$$1 + R \leq C \frac{Y^\alpha}{p}, \quad (4.19)$$

because $\gamma = c_2 q_* Y^{-\alpha}$ and $q \leq q_*$.

We need a bound on the left boundary that does not involve n^k . The identity $1 - |Z|^2 = 4 \operatorname{Re} w/|w + 1|^2$ gives, when $\operatorname{Re} v = -1/2$,

$$1 - |Z_r(v)| \geq \frac{1 - |Z_r^*|}{4(1 + |v|)^2}. \quad (4.20)$$

To check the constant, first use $1 - |Z_r(v)| \geq \xi_r/|w_r(v) + 1|^2$ and $1 - |Z_r^*| \leq 4\xi_r/|w_r + 1|^2$. Then use

$$|w_r(v) + 1| \leq |w_r + 1| + \xi_r |v| \leq (1 + |v|)|w_r + 1|.$$

The coefficients of F have magnitude at most two. Summing independent geometric series over the positive prefix and gap lengths therefore gives, for every point in the open unit polydisk,

$$|F(Z)| \leq 2 \prod_{r=0}^{k-1} \frac{1}{1 - |Z_r|}.$$

It follows from (4.20) that

$$g(v) := \frac{F(Z_0(v), \dots, Z_{k-1}(v))}{(2 + v)^{2k}}$$

has the boundary bounds

$$\begin{aligned} \log|g(v)| &\leq W_* + Ck + C & (\operatorname{Re} v = -1/2), \\ \log|g(v)| &\leq k \log(n/p) + \log \varepsilon & (\operatorname{Re} v = R). \end{aligned} \quad (4.21)$$

On the left, $(1 + |v|)/|2 + v|$ is bounded by an absolute constant. On the right, $|2 + v| \geq 2 + R > 1$. The function g is analytic and bounded on the strip: F is a finite polynomial evaluated in the unit polydisk, and $|2 + v| \geq 3/2$.

For clarity, the strip estimate we use is the following elementary consequence of the maximum principle. If a bounded analytic function on $a \leq \operatorname{Re} v \leq b$ has logarithmic boundary bounds L_- , L_+ , then at a real $t \in [a, b]$ its logarithm is at most $((b - t)L_- + (t - a)L_+)/ (b - a)$. Multiply the function by the exponential of the negative affine interpolant to make both side bounds one. On rectangles of height tending to infinity, apply the maximum principle after multiplying further by $\exp\{\tau(v^2 - \max\{a^2, b^2\})\}$, for $\tau > 0$. This factor has modulus at most one on the vertical sides and tends uniformly to zero on the horizontal sides. Let the height tend to infinity and then τ tend to zero. This proves the estimate, including zeros of the function by continuity.

At $t = 0$ in our strip, the right boundary has weight $1/(2R + 1)$. By (4.11), $\log|g(0)| \geq -E_h - 1 - 2k \log 2$. Substituting (4.21) consequently gives

$$\begin{aligned} \log(1/\varepsilon) &\leq k \log(n/p) + (2R + 1)(E_h + 1 + 2k \log 2) + 2R(W_* + Ck + C) \\ &\leq k \log(n/p) + C(1 + R)(E_h + 4^h A_0). \end{aligned} \quad (4.22)$$

This is the desired quantitative transfer from input coordinates to trace coordinates.

It remains to compare the exponent with the budget in (1.1). Put $t = \log_2 \mu$. The choices in (4.2) satisfy $t + \alpha < 3$, and

$$4^h \leq CY^2, \quad E_h \leq CA_0 Y^t, \quad A_0 \leq Q(1 + \log Y), \quad QY \leq C \log n.$$

Thus (4.19) yields

$$\begin{aligned} (1 + R)E_h &\leq \frac{C}{p}(\log n)Y^{t+\alpha-1}(1 + \log Y) \leq \frac{C}{p}(\log n)Y^2(1 + \log Y), \\ (1 + R)4^h A_0 &\leq \frac{C}{p}(\log n)Y^{1+\alpha}(1 + \log Y) \leq \frac{C}{p}(\log n)Y^2(1 + \log Y). \end{aligned}$$

Also, since $n \geq 2$ and $\log(1/p) \leq 1/p$,

$$k \log(n/p) \leq CY^2(\log n + \log(1/p)) \leq \frac{C}{p}(\log n)Y^2.$$

Using (4.1), we conclude that the right-hand side of (4.22) is at most

$$\frac{C}{p}(\log n)Y^2(1 + \log Y) \leq C_6 p^{-1}(\log n)X(n, p)^2(1 + \log X(n, p))^6 = C_6 B(n, p). \quad (4.23)$$

Taking $\varepsilon = e^{-C_{\text{sep}} B(n, p)}$ with $C_{\text{sep}} > C_6$ contradicts (4.22). Hence (4.15) fails for that ε , proving Theorem 2.2 with the cutoff (4.2).

All constants were fixed in the following order: those of Proposition 3.1; then J_* , H and η ; then C_5, c_2 ; and finally C_{sep} . There is no dependence on the denominator of p or on how close it is to an endpoint. In particular, when $q < 1/n$ the cutoff $q_* = 1/n$ keeps $Y < 2$ and $Q = 1 + \log n$, while $q/q_* \leq 1$ keeps the strip estimate uniform. When p is small, the strip width is controlled by the single factor p^{-1} already present in the budget. The case $p = 1$ will be handled directly in the decoding argument.

5 A uniform decoder

Theorem 2.2 supplies a test for the first incorrect bit of a prefix: no moment system fixing that prefix can reproduce all the low-order trace statistics. To turn this statement into an algorithm, we must also recognize a correct prefix in bounded time. The useful additional fact is that a correct prefix admits a moment system with strict slack, both in its statistical constraints and in its positivity constraint. We first exhibit that slack, then give a rational procedure which is guaranteed to find a feasible system whenever such slack exists.

5.1 Statistical constraints for a prefix

Assume for now that $p < 1$. Choose an integer D and a positive dyadic number $\zeta = 2^{-b_\zeta}$ such that

$$D_* \leq D \leq CY^2, \quad \zeta \leq \frac{1}{4}, \quad 4\zeta < e^{-C_{\text{sep}} B(n, p)}, \quad b_\zeta = O(B(n, p)), \quad (5.1)$$

where C_{sep} is the constant in Theorem 2.2. Integer formulas for these choices will be given below. Let

$$\mathcal{I}_D = \{I \subseteq \{1, \dots, n\} : 1 \leq |I| \leq D\}, \quad N = |\mathcal{I}_D|, \quad T = 512(1 + N)\zeta^{-2}.$$

The number T is an integer. From T independent traces, form the empirical mean $\hat{t}_I$ of the rank observable defining T_I , for every $I \in \mathcal{I}_D$. These are observables of the ordinary trace: if a requested rank is absent, its observable is zero.

For each fixed input x , let $\mathcal{G}$ be the event

$$|\hat{t}_I - T_I(x)| \leq \zeta/8 \quad \text{for every } I \in \mathcal{I}_D. \quad (5.2)$$

Each observable takes values in $\{0, 1\}$, so its empirical mean has variance at most $1/(4T)$. Chebyshev's inequality and a union bound give

$$\mathbb{P}_x(\mathcal{G}^c) \leq \frac{16N}{T\zeta^2} \leq \frac{1}{32}. \quad (5.3)$$

The same event will control all the prefix tests, even though the prefixes are chosen using the data.

For a proposed prefix, use the remaining sign variables and their moment matrix $M(m) = \mathcal{M}_D(m)$ from (2.1). Its nonempty moment coordinates form a real vector m . Consider the rational constraints

$$m \in [-1, 1]^l, \quad M(m) \succeq 0, \quad |\mathcal{E}_m[T_I] - \hat{t}_I| \leq \zeta \quad (I \in \mathcal{I}_D), \quad (5.4)$$

where l is the number of moment coordinates. Substituting the prefix and writing each remaining bit as one half of one plus its sign makes $\mathcal{E}_m[T_I]$ an affine function of m . Its nonconstant coefficients have absolute sum at most 1, by the coefficient bound for T_I in Section 2.

Lemma 5.1. *On $\mathcal{G}$, a prefix whose first incorrect bit is at its last position has no solution to (5.4). Every correct prefix has a moment vector $m^* \in [-1, 1]^l$ such that, writing G for the dimension of its moment matrix,*

$$M(m^*) \succeq (\zeta/8) \text{Id}_G, \quad |\mathcal{E}_{m^*}[T_I] - \hat{t}_I| \leq \zeta/2 \quad (I \in \mathcal{I}_D). \quad (5.5)$$

Proof. If a wrong prefix had a feasible moment vector, then (5.2) and (5.4) would give

$$|\mathcal{E}_m[T_I] - T_I(x)| \leq 9\zeta/8 \quad (I \in \mathcal{I}_D).$$

This contradicts Theorem 2.2 and (5.1).

For a correct prefix, take the moments of the true completion with weight $1 - \zeta/8$, and the moments of independent uniform signs on the remaining positions with weight $\zeta/8$. The moment matrix of the latter distribution is the identity: its (I, J) entry is zero unless $I = J$. The mixture therefore has the stated matrix lower bound, and its coordinates lie in $[-1, 1]$. Both constituent distributions give channel expectations in $[0, 1]$, so mixing changes each $T_I(x)$ by at most $\zeta/8$. Equation (5.2) then bounds the discrepancy from $\hat{t}_I$ by $\zeta/4$, which is stronger than required. $\square$

5.2 Finding a moment system with slack

The next lemma gives the bounded procedure needed for (5.4). It only promises to succeed when there is slack; every reported solution satisfies the exact inequalities. This distinction avoids any need to decide arbitrary instances of semidefinite feasibility. The procedure is a projected relaxation method for inequalities, in the tradition of Agmon and Motzkin–Schoenberg [1, 16]. We give the iteration and its rounding analysis explicitly.

Lemma 5.2 (Rational feasibility with slack). *Let $r \geq 0$ and $D \geq 1$ be integers, and let m consist of the*

$$l = \sum_{j=1}^{\min(2D, r)} \binom{r}{j}$$

nonempty sign moments. Let $M(m)$ be their moment matrix, of dimension

$$G = \sum_{j=0}^{\min(D,r)} \binom{r}{j}.$$

Suppose a finite list of rational affine inequalities $\ell_t(m) \leq b_t$ is given, each with nonconstant coefficient vector of ℓ^1 -norm at most 1, together with a dyadic $0 < \zeta \leq 1/4$.

There is a deterministic rational procedure which either returns a vector satisfying

$$m \in [-1, 1]^l, \quad M(m) \succeq 0, \quad \ell_t(m) \leq b_t \quad \text{for all } t,$$

or reports failure. It returns a vector whenever some $m^* \in [-1, 1]^l$ satisfies

$$M(m^*) \succeq (\zeta/8) \text{Id}_G, \quad \ell_t(m^*) \leq b_t - \zeta/2 \quad \text{for all } t.$$

Put $\lambda = \zeta/(16G)$. The procedure performs at most $\lceil 10(1+l)/\lambda^2 \rceil$ feasibility checks, each taking a number of bit operations polynomial in the list dimensions, the input bit lengths, and $\log(1/\zeta)$. These time bounds hold without the slack assumption.

Proof. If $l = 0$, the matrix is $[1]$ and all inequalities are constant, so direct rational comparisons suffice. Suppose $l > 0$. We describe exact separating directions, then an iteration with bounded precision.

Directions from violated constraints. At a current point $m^{\text{cur}} \in [-1, 1]^l$, first check the affine inequalities and the matrix condition exactly. If all hold, return the current vector. A violated inequality $\ell_t(m^{\text{cur}}) > b_t$ gives the direction a equal to minus its nonconstant coefficient vector. Then $\|a\|_2 \leq 1$; under the slack assumption,

$$\langle m^* - m^{\text{cur}}, a \rangle \geq \zeta/2 \geq \lambda.$$

If instead the matrix condition fails, find a nonzero rational vector v with $v^\top M(m^{\text{cur}})v < 0$. The exact construction of v is given just below. Write the associated affine form as

$$v^\top M(m)v = c + h \cdot m, \quad a = \frac{h}{Gv^\top v}.$$

Each matrix entry is either the constant 1 or one moment coordinate. Consequently

$$\|h\|_1 \leq \|v\|_1^2 \leq G\|v\|_2^2, \quad \|a\|_2 \leq 1.$$

The slack matrix lower bound gives

$$\langle m^* - m^{\text{cur}}, a \rangle = \frac{v^\top (M(m^*) - M(m^{\text{cur}}))v}{Gv^\top v} \geq \frac{\zeta}{8G} \geq \lambda.$$

Thus either violated condition provides a rational direction satisfying

$$\|a\|_2 \leq 1, \quad \langle m^* - m^{\text{cur}}, a \rangle \geq \lambda \tag{5.6}$$

whenever a slack solution exists. If a constant affine inequality is violated, its direction is zero; this cannot happen in the slack case, and in all other cases the iteration is still bounded.

An exact matrix check. For a rational symmetric matrix, eliminate a positive diagonal pivot by taking its Schur complement. Congruence by an invertible triangular matrix shows that the original matrix is positive semidefinite exactly when the positive pivot and this Schur complement are. A negative diagonal supplies a negative unit vector. If a remaining diagonal at i is zero and its entry b in column j is nonzero, let c be the diagonal at j . We may assume $c \geq 0$, since otherwise there is already a negative diagonal. The rational vector

$$y = e_i - \frac{b}{1 + c + |b|} e_j$$

has quadratic form

$$\frac{b^2}{(1 + c + |b|)^2} [-2(1 + c + |b|) + c] < 0.$$

If no negative vector is found, continue with a positive diagonal pivot; when none remain, all entries are zero and the remainder is positive semidefinite.

For completeness, a negative vector of a Schur complement lifts to one of the original matrix using only rational arithmetic. If P is the original block of eliminated pivots and B its block of entries against the remaining indices, the lift of y is

$$v = (-P^{-1}By, y).$$

Its quadratic form is exactly the quadratic form of y for the Schur complement.

These operations have polynomial bit complexity. Indeed, clear denominators in a rational matrix with t -bit entries; even the product of all denominators has at most $G^2 t$ bits. If H is the resulting integer matrix and its entries have at most s bits, the determinant expansion bounds every j -row minor by $j! 2^{js}$. Its bit length is therefore $O(G(s + \log G))$. Schur-complement entries are ratios of minors of the original matrix, and the inverse formula for P gives the same bound for the lift. Fraction-reduced elimination, with rational comparisons at each step, therefore computes the check and the negative vector in polynomially many bit operations. This argument applies however small a positive pivot or a negative quadratic form may be.

Iteration and rounding. Start with $m^{\text{cur}} = 0$. Choose the dyadic grid with mesh

$$\eta = 2^{-g}, \quad g = \left\lceil \log_2 \frac{100(1 + l)}{\lambda^2} \right\rceil.$$

The integer g is computed by rational comparisons with powers of two; thus $\eta \leq \lambda^2/(100(1 + l))$, with $O(1 + \log((1 + l)/\lambda))$ fractional bits. After an unsuccessful check, add λa , clip every coordinate to $[-1, 1]$, and round to the nearest point of this grid in the box. The endpoints of the box belong to the grid. Repeat up to the stated number of checks, reporting failure if none accepts.

Under the slack assumption, (5.6) implies

$$\|m^{\text{cur}} + \lambda a - m^*\|_2^2 \leq \|m^{\text{cur}} - m^*\|_2^2 - \lambda^2.$$

Clipping cannot increase the distance to m^* . Since both the clipped vector and m^* lie in the box, rounding changes the squared distance by at most $4l\eta + l\eta^2 < \lambda^2/2$. Each failed check thus decreases squared distance by at least $\lambda^2/2$. Its initial value is at most l , so acceptance must occur before the cap.

Finally, rounding keeps every iterate on a single grid of the indicated bit length. The exact matrix check, the affine comparisons, the formation and normalization of h , and one update

involve polynomially many rational operations on numbers of polynomial bit length. Temporary denominators within an update therefore have polynomial bit length, and they do not accumulate across updates. The fixed cap and these bounds apply to every rational input list, whether or not it admits a slack solution. $\square$

For a prefix test, write each absolute-value constraint in (5.4) as two affine inequalities and apply Lemma 5.2. Lemma 5.1 verifies exactly the slack hypotheses for a correct prefix. Every acceptance, on any data, gives an actual moment system satisfying (5.4).

The decoder starts with the empty prefix and tests both one-bit extensions. If precisely one accepts, it appends that bit; otherwise it appends 0. It repeats until the prefix has length n . The latter rule specifies a length- n output even for data on which the statistical estimates fail. There are at most $2n$ tests.

5.3 Integer parameters and bit complexity

We now make the choices in (5.1) using only the binary input. This also accounts for the size of every list and every rational calculation in the decoder. Continue to assume $p < 1$, and use q_* , Q , Y from (2.3). Define the positive integers

$$\begin{aligned} u &= 1 + \lceil \log_2 n \rceil, & v &= 1 + \lfloor \log_2(1/q_*) \rfloor, \\ y &= 1 + \lceil u/v \rceil, & w &= 1 + \lceil \log_2 y \rceil, & a_p &= \lceil 1/p \rceil. \end{aligned}$$

All logarithms appearing in these integer formulas mean integer length comparisons. In particular, the integer part of a logarithm of a rational number is found by comparing its numerator with power-of-two multiples of its denominator. The rational maximum defining q_* is also found by an integer comparison.

The elementary comparisons

$$u \asymp \log n, \quad v \asymp Q, \quad Y \leq y \leq CY, \quad w \asymp 1 + \log Y, \quad p^{-1} \leq a_p \leq 2p^{-1}$$

have absolute constants. For the asserted lower bound on y , use $u \geq (\log n)/\log 2$ and $v \leq Q/\log 2$. Together with $Y \asymp X(n, p)$, these comparisons give

$$B(n, p) \leq a_p u y^2 w^6 \leq C B(n, p). \quad (5.7)$$

Choose fixed positive integers K_D, K_m , sufficiently large for the constants in Theorem 2.2, and set

$$D = K_D y^2, \quad b_\zeta = K_m a_p u y^2 w^6, \quad \zeta = 2^{-b_\zeta}. \quad (5.8)$$

Since $D_* \leq CY^2$ and $B(n, p) \geq \log 2$, these choices ensure all of (5.1). The integers D, b_ζ , in binary, are computable in time polynomial in the input bit lengths. Writing the denominator 2^{b_ζ} subsequently costs $O(b_\zeta)$ bits, which will be included in the bounds below.

We next describe how to form the rational constraints. For $I = \{l_1 < \dots < l_k\}$ and $i_1 < \dots < i_k$, the coefficient of $X_{i_1} \dots X_{i_k}$ in T_I is

$$\binom{i_1 - 1}{l_1 - 1} \prod_{j=1}^{k-1} \binom{i_{j+1} - i_j - 1}{l_{j+1} - l_j - 1} p^{l_k} q^{i_k - l_k}, \quad (5.9)$$

when all the gap counts are possible, and is zero otherwise. Indeed, the selected positions must be retained, and the prefix and successive open gaps must contain the indicated numbers of additional

retentions. Positions after i_k are unrestricted. Formula (5.9) follows by multiplying these independent binomial probabilities.

If the input representation is $p = a/b$, every denominator in (5.9) divides b^n . Binomial coefficients have $O(n)$ bits. Sign expansion adds a denominator dividing 2^D , and empirical means have denominator T . Consequently the constraint coefficients have polynomial bit length in $n + L + D + \log T$. Enumerating the rank and input tuples, then expanding the resulting bit monomials and substituting the prefix, constructs them directly. No list of complete trace masks is required.

All these enumerations fit the permitted resource scale. From $D = O(Y^2)$, $Y \asymp X(n, p)$, and (1.1),

$$D \log n = O(B(n, p)). \quad (5.10)$$

The rank list, every matrix basis, and every moment list have size at most $(n+1)^{2D} = \exp(O(B(n, p)))$. Input tuple lists satisfy the same bound. A sign expansion has at most 2^D terms; products of these enumeration bounds remain $\exp(O(B(n, p)))$. Subsets larger than the available number of positions are simply absent, so this statement includes $D > n$ and prefixes with very few remaining positions.

By (5.1), the trace count T is $\exp(O(B(n, p)))$. For every prefix, the iteration cap in Lemma 5.2 is bounded by

$$2560(1+l)G^2\zeta^{-2} + 1 = \exp(O(B(n, p))).$$

The bit length of its grid is $O(b_\zeta + \log G + \log(1+l)) = O(B(n, p))$. The preceding coefficient bounds and that lemma therefore bound all computation by a fixed polynomial in

$$n + L + \exp(O(B(n, p))). \quad (5.11)$$

In particular, this includes the construction of parameters and constraints, exact rational arithmetic, and the at most $2n$ prefix tests.

Proof of Theorem 1.1. If $p = 1$, request one trace and output it. For the universal halting specification, output 0^n if the supplied well-formed trace has a different length. The actual channel always supplies the input itself, and this branch takes polynomially many bit operations in $n + L$.

For $p < 1$, use (5.8), request the specified T traces, and compute their empirical statistics. Run the prefix rule specified above. This defines the machine on every stream of well-formed traces.

On $\mathcal{G}$, induction on the prefix length shows that precisely the correct extension accepts: the incorrect extension is infeasible by Lemma 5.1, and the correct one has the slack that guarantees acceptance. Thus the output is x on $\mathcal{G}$, whose probability is at least $31/32 > 2/3$ for each fixed x .

Each supplied trace has at most n bits and one delimiter. Reading them, and computing each of the finitely many rank observables, costs polynomially many bit operations in $n + T + N$; the empirical counters have $O(\log T)$ bits. All loops, including all feasibility tests, have the input-dependent caps already given. Thus (5.11) is a bound for every well-formed trace stream, independently of the success event. The machine can ignore its internal random bits.

To be explicit about the computational model, the described finite lists may be stored sequentially on tapes; replacing an indexed lookup by a scan incurs at most a polynomial overhead in their total encoded length. Integer addition, multiplication, division, and fraction reduction have elementary polynomial-time multitape implementations. The bit-length and loop bounds above therefore give absolute constants K, d for which the trace count is at most $\exp(KB(n, p))$ and the total bit cost is at most $(n + L + \exp(KB(n, p)))^d$, after enlarging K, d to absorb fixed factors. This enlargement is uniform because $B(n, p) \geq \log 2$. Choose absolute C, c large enough, with $C \geq K$ and $c \geq d$. Both the sample bound $M_C(n, p)$ and the required runtime $(n + L + M_C(n, p))^c$ follow. All constants used in parameter selection are fixed integers in a single finite program. $\square$

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